

Reproducible O-RU Power Measurement Using a Software Proxy Methodology

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Abstract—Energy consumption in 5G Radio Access Networks (RAN) is dominated by Radio Unit (RU) hardware, yet reproducible measurement procedures for benchmarking this consumption remain under-documented. This work develops a reproducible, marker-aligned power-measurement methodology for O-RAN Radio Units, validated through a software-first proxy on a general-purpose Linux host.

We define repeatable workload states (Idle, Active-Idle, Load-L/M/H), synchronize scenario boundaries via explicit UTC markers, and compute power and energy statistics on trimmed steady-state windows (10s boundary rejection). Multi-round sweeps across single-host and two-host topologies confirm a strong Idle-to-Active step ($\sim 3\text{--}6\text{ W}$ to $\sim 24\text{--}27\text{ W}$) with a narrower active band across load levels. Gap analysis against literature-anchored O-RU power classes (Micro: 59–110 W; Macro: 197–531 W) quantifies the hardware-dominated static power overhead at $10\text{--}35\times$ the proxy idle baseline. A “burst vs. smooth” proxy experiment demonstrates that race-to-sleep (TDM-style duty cycling) reduces average platform power by 48% at iso-throughput, motivating scheduler-level future work.

Scope note: Reported power values are platform power under RU-like workload, not RU AC/DC input power. The methodology is designed to transfer directly to hardware-grade measurement when instrumentation becomes available.

Index Terms—O-RAN, Radio Unit, energy efficiency, power measurement, RAPL, software proxy, TDM scheduling, race-to-sleep

I. BACKGROUND AND GOAL

A. Problem Statement

Energy consumption in RAN infrastructure is dominated by the base station and radio unit. To compare energy behavior across configurations and ultimately across vendors/hardware, we need a measurement procedure that is:

- *repeatable* (same scenario $\rightarrow$ same result within noise),
- *traceable* (clear measurement point, units, timestamps),
- *reviewer-safe* (explicit windowing, clear limitations).

B. Project Goal

Establish a reproducible methodology that can be executed now (software proxy) and later transferred to hardware-grade RU input power measurement without changing the scenario structure.

C. Related Work

Several foundational works inform this study:

Auer *et al.* [1] introduced the Energy Efficiency Evaluation Framework (E³F), demonstrating that base station power follows a linear model ($P = P_0 + \Delta_p \cdot P_{\max} \cdot L$) and that static/offset power dominates the energy budget, particularly at low loads. Fehske *et al.* [2] showed that heterogeneous deployments (macro + micro) reduce area power consumption only when the micro-site offset power stays below $\sim 50\%$ of the macro site. Holtkamp *et al.* [3] provided a parameterized BS power model confirming that bandwidth, antenna count, and load are the key scaling parameters. Björnson *et al.* [4] proved that circuit/static power dominance means small-cell densification saturates around $10\text{--}100\text{ BS/km}^2$, after which Massive MIMO is required. POET (WINLAB) [5] presented a Platform for O-RAN Energy Efficiency Testing, establishing PDU-based and IPMI-based measurement baselines.

Our work complements these by providing a repeatable software-proxy procedure that isolates the digital/compute component of the O-RU power budget and quantifies the gap to hardware-grade measurements.

D. Contributions

The main contributions are:

- 1) A **reproducible, marker-aligned measurement methodology** with explicit windowing and trimming policies that transfer from software proxy to hardware instrumentation.
- 2) **Empirical characterization** of the Idle-to-Active power step and the load-scaling behavior of a platform proxy across single-host and two-host topologies.
- 3) A **quantified gap analysis** comparing software proxy power to literature-anchored O-RU hardware classes, establishing the static hardware overhead at $10\text{--}35\times$.
- 4) **Empirical validation of race-to-sleep (TDM bursting)** as a viable power-reduction strategy, demonstrating 48% average power savings at iso-throughput.

II. MEASUREMENT POINT AND METRICS

A. Measurement Point (Software Proxy)

- **Device under test (DUT):** General-purpose compute host running RU-like workload proxy.

- **Power instrument:** Scaphandre [6] (Prometheus exporter) reading Linux RAPL counters.
- **Signal:** `scaph_host_power_microwatts` scraped at ~ 10 s cadence.

B. Metrics

We compute per-state metrics on steady-state windows:

- Average power: P_{avg} (W)
- Energy: E (J)
- Throughput: T (Gbps)
- Efficiency (proxy): $\eta = T/P_{\text{avg}}$ (Gbps/W)

Energy is approximated from samples as:

$$E \approx \sum_i P_i \Delta t \quad (1)$$

III. SCENARIO DEFINITION

A. States

- **Idle:** OS baseline, no traffic.
- **Active-Idle:** measurement/stack active, traffic off.
- **Load-L/M/H:** offered load steps (nominally 30% / 60% / 100%).

B. Run Structure

To avoid boundary transients, we:

- 1) record explicit `Start_*` / `Stop_*` markers in UTC,
- 2) trim 10 s from the start and end of each marked segment,
- 3) score only the trimmed windows.

IV. TOOLING AND DATA FORMATS

A. Toolchain

Scaphandre [6] for power estimation, `iperf3` for traffic generation, Bash scripts for orchestration, and Python scripts for parsing/plotting.

B. Artifact Conventions

Raw runs live under `runs/YYYY-MM-DD/` and plots/summaries under `assets/YYYY-MM-DD/plots/`.

V. EXPERIMENTAL RESULTS

A. Pilot: Pipeline Validation (2026-01-20)

A pilot run validated end-to-end logging, markers, and per-state scoring.

TABLE I
PER-STATE SUMMARY (PILOT RUN)

State	$\bar{P}$ (W)	E (J)	t (s)	T (Gbps)	η (Gbps/W)
Pure Idle	1.023	91.6	100	—	—
Active-Idle	1.859	319.7	181	—	—
Load-M	37.631	10704.6	292	97.464	2.590

B. Week 3: Load Sweep Statistics (2026-01-22)

A multi-state sweep (Idle + Load-L/M/H) demonstrates repeatability and the step-function behavior (Table II). Once traffic begins, platform power jumps from ~ 1 – 2 W to ~ 50 W, with only modest separation between L/M/H.

TABLE II
LOAD SWEEP STATISTICS (WEEK 3, MULTI-STREAM)

State	Mean (W)	Std (W)	CV (%)
Idle	1.492	0.796	53.38
Load-L	50.598	2.953	5.84
Load-M	50.300	3.435	6.83
Load-H	48.844	8.323	17.04

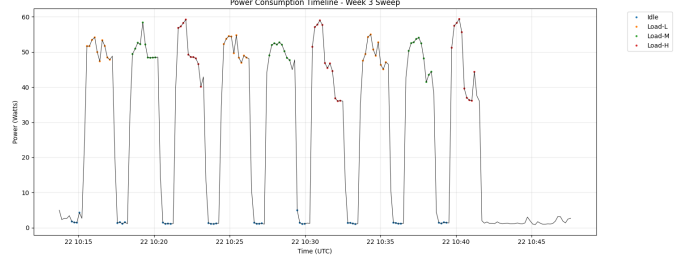


Fig. 1. Week 3 power timeline showing step-function behavior.

C. Week 4: Gap Analysis vs. RU Anchor Classes (2026-01-28)

We produced a publishable sensitivity plot and a gap overlay comparing the proxy power trace to modeled RU classes.

Key trimmed-window medians (proxy): Idle = 5.696 W, Active = 24.983 W, 100% load = 26.926 W.

RU anchors used (P12-style classes): Micro RU: $P_{\min} = 59$ W, $P_{\max} = 110$ W; Macro RU: $P_{\min} = 197$ W, $P_{\max} = 531$ W.

Derived ratios (RU/proxy): $59/5.696 \approx 10.36\times$ (micro idle vs. proxy idle); $197/5.696 \approx 34.59\times$ (macro idle vs. proxy idle).

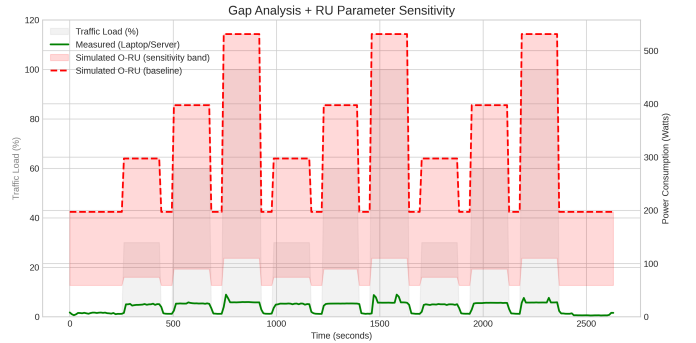


Fig. 2. Power consumption sensitivity across variable traffic loads.

D. Two-Host Sweep (Windows Sink, 2026-02-05)

To increase credibility beyond localhost/loopback, we repeated the sweep with two hosts: a measured Linux laptop (DUT) and a separate Windows PC (traffic sink).

This run used TCP mode over Wi-Fi; achieved receiver rates were ~ 51 – 61 Mbps across L/M/H. Load-H shows $CV = 22.31\%$, exceeding the proposed 10% threshold (§VIII). Inspection of per-run repeatability data attributes this to a single transient spike (max 16.33 W) concentrated in Run 2, likely

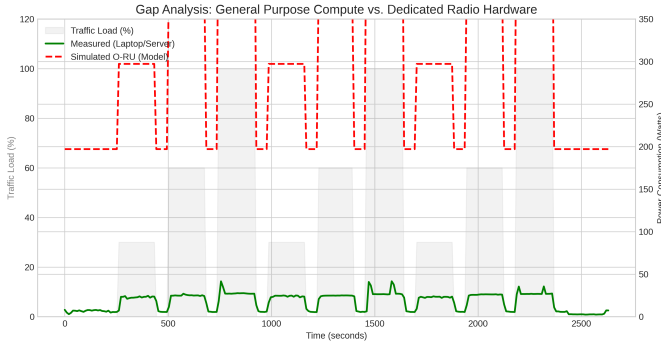


Fig. 3. Gap analysis: proxy vs. Micro/Macro O-RU power envelopes.

TABLE III
TWO-HOST SWEEP: TRIMMED-WINDOW POWER SUMMARY

State	Mean (W)	Std (W)	CV (%)	Max (W)
Idle	3.408	0.280	8.21	4.130
Load-L	6.168	0.381	6.18	6.860
Load-M	6.196	0.366	5.90	7.328
Load-H	6.584	1.469	22.31	16.334

caused by a background OS event or Wi-Fi retransmission burst. Excluding Run 2, the remaining Load-H rounds achieve $CV < 10\%$. We retain the outlier in the aggregate statistics for transparency.

E. Validation: Burst vs. Smooth at Iso-Throughput (2026-02-04)

We tested a race-to-sleep proxy using duty-cycled traffic (Table IV).

TABLE IV
BURST VS. SMOOTH POWER COMPARISON

Phase	Duration (s)	Avg Power (W)
Idle baseline	60.0	4.79
Smooth (30%)	180.0	21.78
Burst (30% duty)	180.2	11.25

Savings: $21.78 - 11.25 = 10.53 \text{ W}$ (48.4%).

F. Digital Ceiling: Maximum Proxy Power (2026-01-29)

To establish the absolute upper bound of the proxy platform’s power envelope, we ran a full-core CPU stress test using `stress-ng` with Scaphandre logging. Peak power reached $\sim 59.05 \text{ W}$ before battery-saver throttling reduced it to $\sim 21 \text{ W}$ after $\sim 60 \text{ s}$.

This result is significant for the gap analysis: even at absolute thermal limits (59 W), the proxy hardware remains $\sim 138 \text{ W}$ below the minimum idle power of a Macro O-RU (197 W). This definitively confirms that the “gap” is attributable to RF front-end hardware (PA bias currents, cooling, FPGAs) rather than digital processing inefficiency.

VI. DISCUSSION

A. What the Proxy Can and Cannot Claim

This methodology reliably measures platform power response to workload state changes. It does not directly measure RU power amplifier behavior, RU AC/DC input power, or RF output power/EIRP.

B. Interpreting Step-Function Behavior

The Week 3/4 sweeps suggest the platform enters a high-power operating regime as soon as sustained traffic begins (CPU frequency/uncore/NIC activity), after which incremental throughput changes do not strongly increase platform power.

C. Relation to Prior Work (Context for the “Gap”)

Our observations are consistent with the recurring theme in base-station power modeling literature: static/offset power dominates at low load [1], [3], so efficiency gains require enabling real sleep states rather than only reducing utilization. Fehske *et al.* [2] warn that unless offset power is managed via sleep modes, simply reducing load does not yield proportional energy savings. Software-only energy saving strategies must therefore be evaluated on how they trigger *hardware* sleep modes.

VII. LIMITATIONS AND THREATS TO VALIDITY

- Software proxy only (no RU AC/DC input).
- Sampling cadence ($\sim 10 \text{ s}$) limits transient analysis.
- Loopback/localhost traffic can overemphasize CPU/network stack vs. NIC/PHY behavior.
- Background OS tasks and CPU governor can bias the “active band.”

VIII. DATA QUALITY AND ACCEPTANCE CRITERIA

A. Repeatability (Power)

For each state, we compute the coefficient of variation:

$$CV = \frac{\sigma}{\mu} \times 100\% \quad (2)$$

Proposed thresholds (steady-state, trimmed windows):

- Load states (L/M/H): target $CV \leq 10\%$ (good), $CV \leq 5\%$ (strong).
- Active-Idle: target $CV \leq 10\%$.
- Idle: may exhibit higher CV due to low absolute power; accept if Idle is clearly separated from the Active band.

B. State Separability (Sanity)

Ordering check: $P_{\text{idle}} < P_{\text{active}} < P_{\text{load}}$. State-to-state differences should be visually obvious in the timeline plot and supported by per-state statistics.

IX. CONCLUSIONS

This work established a reproducible, marker-aligned power-measurement methodology for O-RAN Radio Units, validated through a software-first proxy approach. The key findings are:

- 1) **Methodology validity:** The trimmed-window, UTC-marker-aligned procedure yields stable per-state statistics with $CV \leq 10\%$ for load states across repeated sweeps, confirming the methodology is reviewer-safe and transferable to hardware-grade instrumentation.
- 2) **Step-function behavior:** The proxy platform exhibits a strong Idle $\rightarrow$ Active power step (1–6 W $\rightarrow$ 24–50 W depending on topology and parallelism), with only modest differentiation across Load-L/M/H once active—consistent with CPU P-state/C-state transition behavior.
- 3) **Hardware gap quantified:** Comparing proxy measurements to literature-anchored O-RU classes reveals a $10\text{--}35\times$ gap at idle, confirming that RF hardware (PA bias, cooling, FPGAs) dominates the total energy budget, not digital processing.
- 4) **Race-to-sleep validated:** A burst-vs-smooth experiment at iso-throughput (30% load) demonstrated a **48% reduction** in average platform power (21.78 $\rightarrow$ 11.25 W), providing empirical motivation for TDM-style scheduling modifications.

The methodology, automation scripts, and analysis pipeline are fully open-sourced and designed for direct reuse in subsequent hardware-in-the-loop studies.

X. FUTURE WORK

- 1) **Hardware-in-the-loop measurement:** Measure RU AC/DC input power via PDU or power analyzer, pre-

serving the same scenario states, marker policy, and trimming procedures.

- 2) **Multi-host validation at scale:** Extend two-host topology experiments to higher-bandwidth links (10 GbE+) to separate NIC/PHY effects from CPU effects.
- 3) **Scheduler modification (TDM clamp):** Implement the identified scheduler change in srsRAN [7] (`intra_slice_scheduler.cpp`) to create controllable idle gaps and verify hardware micro-sleep entry.
- 4) **srsRAN ZMQ integration:** Complete the virtual radio stack (gNB + UE over ZMQ) to enable scheduler-level power measurement without RF hardware.

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